LASIUS HEXAPOD — TIBIA (1:1 SCALE TEMPLATE)

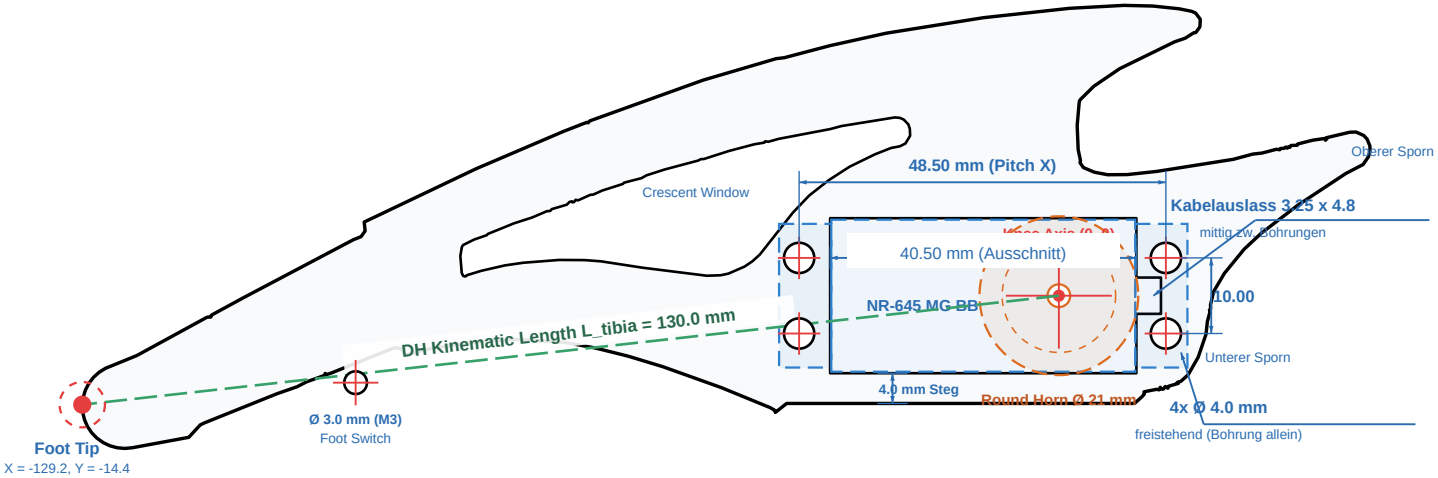
True-to-scale 1:1 inspection & drill template for lower limb / sickle claw

Plate thickness: 7.0 mm | Knee servo NR-645 MG BB (Ø 4.0 mm through-holes) | Kinematics L_DH = 130.0 mm

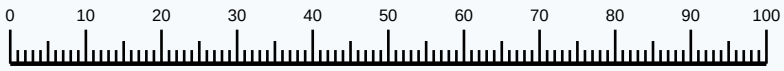
PRINTER SETTING: 100% (Actual Size)

DO NOT use "Fit to page" scaling!

Verify calibration ruler at bottom left with calipers before drilling.



1:1 CALIBRATION CHECK



100.0 mm reference span (measure with calipers)

50.0 mm x 15.0 mm

Check Box

Check Criteria:

- Span 0–100 mm = exactly 100.0 mm
- Box width = exactly 50.0 mm
- Tibia plate thickness = 7.0 mm

PART SPECIFICATIONS & DRILL INSTRUCTIONS

Position Reference:

- Knee pivot axis = Origin (0.0, 0.0) mm
- Foot tip = (-129.15, -14.41) mm | Kinematics: 129.95 mm (nominal 130 mm)
- Foot switch / stopper hole = (-93.00, -11.50) mm | Ø 3.0 mm (M3)

Servo Cutout & Mounting (NR-645 MG BB):

- Window cutout: 40.5 mm x 20.5 mm (X: -30.25..10.25, Y: -10.25..10.25)
- Cable notch: 3.25 mm x 4.80 mm (centered between mounting holes)
- 4x Servo mounting holes: Pitch X = 48.50 mm, Pitch Y = 10.00 mm
- Hole: Ø 4.0 mm through (smooth, without countersink)